

Preliminary Master Use Case List (3/12/26)

CSE416, S01 – Spring 2026

Project goals

- Understand the implications of the potential Supreme Court dismissal of Section 2 of the Voting Rights Act (VRA)
- Assess the impact on a pre-clearance state and a non-preclearance state

Approach

- Develop an estimate of the probability distribution of minority representation under the existing VRA
- Compare with the estimated probability distribution of minority representation under a restructured VRA (i.e., race-blind districting)
- Apply data analysis techniques to the available data
- Generate data tables and visualizations that summarize the results of the analysis

Terminology

- Population – the population of a region (e.g., state, precinct, etc.) refers to the total population as defined by the US Census Bureau. Some calculations might refer to the voting age population (VAP) or citizen voting age population (CVAP). You can use any measure of population, but it should be done consistently throughout the application. Ideal district population is the total population of the state divided by the number of districts in the state.
- Ensemble – a collection of district plans generated on the SeaWulf. A race-blind ensemble contains random district plan, each of which is random and a subset of all of the possible graph partitions that are constrained by the limits on population equality. A VRA-constrained ensemble is further constrained by VRA compliance. A race-blind ensemble is generated with the basic GerryChain software while a VRA-constrained ensemble is created with the VRA constrained GerryChain software.
- Feasible demographic group – a racial/language group for which the state population is at least 400,000. If your state is primarily white (e.g., Iowa), you can lower this threshold to 200,000.
- Demographic group definitions – use the same ones as in the Becker paper, namely every person who identified as Hispanic/Latino on the census or ACS is classified as Latino. The term Black is used for non-Hispanic respondents who selected Black as their single racial category, and White similarly.
 - The Becker paper refers to a “candidate of choice”. This should be replaced by a “party of choice.” For each racial group, the party with the highest average estimated vote total for the 2024 Presidential election is identified as the group’s “party of choice,” with the highest EI frequency as the confidence score.

This draft of the use case list is meant to help you think about requirements. You should think about additional use cases that serve the project goals and also think about more detail for each use case.

General GUI (13 required) – Seongwoo Hong, Minhaj Alok

GUI-1. Select state to display (required) (SD) – Seongwoo Hong (Build 1)

GUI-2. Display the current district plan when state is selected (required) (SD) – Seongwoo Hong (Build 1)

GUI-3. State data summary (required) (SD) – Seongwoo Hong (Build 1)

GUI-4. Display demographic heat map by precinct (required) (SD) – Seongwoo Hong (Build 2)

~~GUI-5. Display demographic heat map by census block (preferred) (SD)~~

GUI-6. Display Congressional representation table (required) (SD) – Seongwoo Hong (Build 1)

GUI-7. Highlight district (preferred) – Seongwoo Hong (Build 2)

GUI-8. Compare two district plans on the map (preferred) (SD) – Seongwoo Hong (Build 3)

GUI-9. Display Gingles analysis results (required) (SD) – Seongwoo Hong, Minhaj Alok (Build 1)

GUI-10. Display the Gingles 2/3 analysis data in a tabular display (preferred) – Seongwoo Hong (Build 1)

GUI-11. Highlight a Gingles 2/3 table row (preferred) – Seongwoo Hong (Build 2)

GUI-12. Display candidate results of Ecological Inference (EI) analysis (required) (SD) – Seongwoo Hong, Minhaj Alok (Build 1)

~~GUI-13. Display EI precinct results in a bar chart (preferred)~~

GUI-14. Display EI precinct results in a choropleth maps (preferred) – Seongwoo Hong (Build 3)

GUI-15. Display EI KDE results (preferred) – Seongwoo Hong, Minhaj Alok (Build 1)

GUI-16. Display ensemble splits in a bar chart (required) (SD) – Seongwoo Hong, Minhaj Alok (Build 1)

GUI-17. Display box & whisker data (required) (SD) – Seongwoo Hong, Minhaj Alok (Build 1)

~~GUI-18. Display vote share vs seat share curve (preferred)~~

GUI-19. Display an “interesting” district plan (preferred) – Seongwoo Hong, Minhaj Alok (Build 1)

GUI-20. Display VRA impact threshold table (required) – Seongwoo Hong (Build 3)

GUI-21. Display minority effectiveness box & whisker data (required) – Seongwoo Hong, Minhaj Alok (Build 3)

GUI-22. Display minority effectiveness ensemble histogram (required) – Seongwoo Hong, Minhaj Alok (Build 3)

GUI-23. Display VRA impact threshold table (required) – Seongwoo Hong (Build 3)

GUI-24. Reset page (preferred) – Seongwoo Hong (Build 1)

Preprocessing (10 required) – Yize Dai

Prepro-1. Integrate multiple data sources (required) (AD)

Prepro-2. Identify precinct neighbors (required) (AD)

Prepro-3. Integrate enacted plan with dataset (required)

Prepro-4. Store preprocessed data (required)

Prepro-5. Store SeaWulf data (required)

Prepro-6. Generate data files required for SeaWulf processing (required) (AD)

Prepro-7. Gingles 2/3 precinct analysis (required) (AD)

Prepro-8. Gingles 2/3 non-linear regression analysis (required) (AD)

Prepro-9. Use the PyEI MGCG software to calculate Ecological Inference data (required)

~~Prepro-10. Calculate the vote share vs seat share curve data (preferred)~~

Prepro-11. Calculate Box & Whisker Data for Enacted Plan (required)

SeaWulf (10 required) – Brandon Koo

SeaWulf-1. Server dispatcher (required)

SeaWulf-2. Run MGCG ReCom algorithm on the SeaWulf (required) (AD)

SeaWulf-3. Run MGCG VRA Constrained ReCom algorithm on the SeaWulf (required) (AD)

SeaWulf-4. Coordinate/aggregate SeaWulf core generated data (required) (AD)
SeaWulf-5. Calculate election winners (required) (AD)
SeaWulf-6. Calculate minority effectiveness score per random district (required)
SeaWulf-7. Calculate minority population percentage per random district (required)
SeaWulf-8. Calculate the Republican/Democratic split for each random district plan (required) (AD)
SeaWulf-9. Identify and store additional random district plans of note (preferred)
SeaWulf-10. Calculate ensemble measures (required) (AD)
SeaWulf-11. Calculate box & whisker data (required) (AD)
SeaWulf-12. Run on multiple SeaWulf nodes (preferred)
SeaWulf-13. Python profiler (preferred)

Server Processing (0 required) – Seongwoo Hong

All server processing addressed in GUI use cases.